

Mean-field Theory of High-Dimensional Bayesian Inference

Part 1: Random Matrix Theory & BBP Phase Transition

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Mathematical Methods for High-dimensional Data
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Roadmap (Lecture 1)

- Motivation from inference: low-rank signals in noise (PCA/denoising)
- GOE & empirical spectral density (ESD)
- Resolvent and Stieltjes transform; inversion formula
- Semicircle law: derivation by the cavity method

Remark (Refs)

A first course in random matrix theory (Bouchaud and Potters)

Introduction to Random Matrices - Theory and Practice (Livan, Novaes and Vivo)

Why Random Matrices for Denoising?

- Problem: infer hidden **low-rank structure** in noisy data (e.g. PCA):

$$\text{Data} = \text{Signal} + \text{Noise} \quad \Rightarrow \quad \mathbf{Y} = \tau \mathbf{x}\mathbf{x}^\top + \mathbf{H} \in \mathbb{R}^{n \times n}$$

- RMT provides a **null model** for the noise-only spectrum
- Later: add a rank-one *spike* and study detectability $\rightarrow$ BBP transition

Null Noise Model: the Gaussian Orthogonal Ensemble (GOE)

Definition (GOE)

A random $n \times n$ symmetric matrix $\mathbf{H} = \mathbf{H}^\top$ belongs to the GOE if its upper-triangular elements are independent with Gaussian law

$$(\mathbf{H})_{ij} = H_{ij} = H_{ji} \sim \begin{cases} \mathcal{N}(0, 2/n) & i = j \\ \mathcal{N}(0, 1/n) & i < j \end{cases}$$

Equivalently, its joint density reads

$$P(\mathbf{H}) \propto \exp\left(-\frac{n}{4} \text{Tr}(\mathbf{H}^2)\right)$$

- $\text{Var}(H_{ij}) = \mathcal{O}(\frac{1}{n})$ to keep eigenvalues/spectral norm $\mathcal{O}(1)$ when $n \gg 1$
- Rotational invariance: $P(\mathbf{H}) = P(\mathbf{O}\mathbf{H}\mathbf{O}^\top) \forall$ orthogonal $\mathbf{O}\mathbf{O}^\top = \mathbf{O}^\top\mathbf{O} = \mathbf{I}_n$

Empirical Spectral Density (ESD)

- (Real) Eigenvalues $\{\lambda_i\}_{i=1}^n$ of $\mathbf{H}$
- **ESD:**

$$\rho_{\mathbf{H}}(\lambda) = \rho_{\mathbf{H}_n}(\lambda) = \frac{1}{n} \sum_{i=1}^n \delta(\lambda - \lambda_i)$$

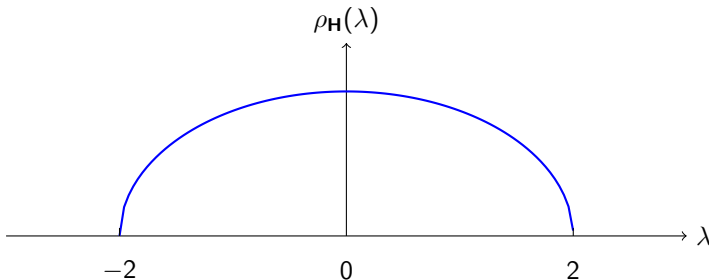
- Goal: describe the **limit** (in distribution) of $\rho_{\mathbf{H}_n}$ as $n \rightarrow \infty$

Wigner's Semicircle Law

Theorem (Wigner's Semicircle Law)

The ESD of a GOE matrix a.s. weakly converges as $n \rightarrow \infty$ to the semicircle law

$$\lim_{n \rightarrow \infty} \rho_{\mathbf{H}}(\lambda) = \rho_{\text{sc}}(\lambda) = \begin{cases} \frac{1}{2\pi} \sqrt{4 - \lambda^2} & |\lambda| \leq 2 \\ 0 & |\lambda| > 2 \end{cases} \quad \text{weakly (i.e., in distribution) a.s.}$$



Definition (Wishart matrix)

Let $\mathbf{X}$ be an $n \times p$ random matrix with i.i.d. entries $X_{ij} \sim \mathcal{N}(0, 1)$, where n is the number of samples, p is the number of features for each sample. The $p \times p$ sample-covariance matrix

$$\mathbf{W} = \frac{1}{n} \mathbf{X}^\top \mathbf{X}$$

is called a **Wishart matrix** (or belongs to the Wishart ensemble).

- $\mathbf{W}$ is symmetric and positive semi-definite
- Eigenvalues $\lambda_i(\mathbf{W}) \geq 0$ represent variances along principal axes (eigenvectors)
- When p is fixed and $n \rightarrow \infty$, $\mathbf{W} \rightarrow \mathbf{I}_p$ (law of large numbers)
- Interesting high-dimensional regime: $n, p \rightarrow \infty$ with $p/n \rightarrow \gamma > 0$

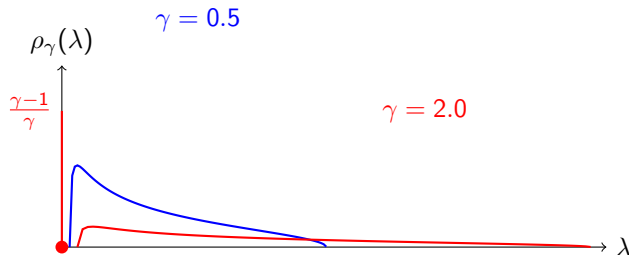
Covariance of Noise: Wishart Matrices and Marcenko-Pastur Law

Theorem (Marcenko-Pastur Law)

As $n, p \rightarrow \infty$ with $p/n \rightarrow \gamma \in (0, \infty)$, the ESD of $\mathbf{W}$ a.s. weakly converges to the Marcenko-Pastur law

$$\lim_{n, p \rightarrow \infty} \rho_{\mathbf{W}}(\lambda) = \rho_{\gamma}(\lambda) = (1 - \gamma^{-1}) \mathbf{1}_{\gamma > 1} \delta(\lambda) + \begin{cases} \frac{1}{2\pi\gamma\lambda} \sqrt{(\lambda_+ - \lambda)(\lambda - \lambda_-)} & \lambda \in [\lambda_-, \lambda_+] \\ 0 & \text{otherwise} \end{cases} \quad \text{a.s.}$$

where $\lambda_{\pm} = (1 \pm \sqrt{\gamma})^2$.



Resolvent & Stieltjes Transform

- **Resolvent:** For $\mathbf{H} = \mathbf{O}\mathbf{\Lambda}\mathbf{O}^\top$,

$$\mathbf{G}_{\mathbf{H}}(z) = (z\mathbf{I}_n - \mathbf{H})^{-1} = \mathbf{O} \text{diag}\left(\frac{1}{z - \lambda_i(\mathbf{H})}\right)_i \mathbf{O}^\top, \quad z \in \mathbb{C} \setminus \{\lambda_i\}_i$$

- **Stieltjes transform:**

$$g_{\mathbf{H}}(z) = \frac{1}{n} \text{Tr } \mathbf{G}_{\mathbf{H}}(z) = \frac{1}{n} \sum_{i \leq n} \frac{1}{z - \lambda_i} = \int d\lambda \frac{\rho_{\mathbf{H}}(\lambda)}{z - \lambda} \quad (\sim 1/z \text{ as } |z| \rightarrow \infty)$$

Remark

We always consider $n \gg 1$, so in many expressions $\lim_{n \rightarrow \infty}$ is implicit

- Moment generating function: $\left. \frac{\partial}{\partial (1/z)^{k+1}} g_{\mathbf{H}}(z) \right|_{z=\infty} = (k+1)! \int \lambda^k \rho_{\mathbf{H}}(\lambda) d\lambda$

Theorem (Stieltjes inversion formula)

$$\rho_{\mathbf{H}}(\lambda) = \lim_{\varepsilon \downarrow 0} \frac{1}{\pi} \text{Im } g_{\mathbf{H}}(\lambda - i\varepsilon)$$

Stieltjes Transform of Null Models

The Stieltjes transforms of the semicircle law and Marcenko-Pastur respectively read

$$1 - zg_{\text{sc}}(z) + g_{\text{sc}}(z)^2 = 0 \Rightarrow g_{\text{sc}}(z) = \frac{z \pm \sqrt{z^2 - 4}}{2}$$

$$g_{\gamma}(z) = \frac{z - 1 + \gamma \pm \sqrt{(z - 1 + \gamma)^2 - 4\gamma z}}{2\gamma z}.$$

The spectral density is recovered via Stieltjes inversion (TO DO).

Stieltjes Transform of GOE via the Cavity Method

Theorem (Schur complement formula)

Let $\mathbf{M} \in \mathbb{R}^{n \times n}$ be invertible, divided as

$$\mathbf{M} = \begin{pmatrix} \mathbf{M}_{11} & \mathbf{M}_{12} \\ \mathbf{M}_{21} & \mathbf{M}_{22} \end{pmatrix} \quad \text{and} \quad \mathbf{M}^{-1} = \begin{pmatrix} \mathbf{M}_{11}^{-1} & \mathbf{M}_{12}^{-1} \\ \mathbf{M}_{21}^{-1} & \mathbf{M}_{22}^{-1} \end{pmatrix},$$

where $\mathbf{M}_{11} \in \mathbb{R}^{t \times t}$, $\mathbf{M}_{12} \in \mathbb{R}^{t \times (n-t)}$, $\mathbf{M}_{21} \in \mathbb{R}^{(n-t) \times t}$, $\mathbf{M}_{22} \in \mathbb{R}^{(n-t) \times (n-t)}$, and $\mathbf{M}_{22}$ is invertible. The integer t can take any values from 1 to $n-1$. Then

$$(\mathbf{M}_{11}^{-1})^{-1} = \mathbf{M}_{11} - \mathbf{M}_{12} \mathbf{M}_{22}^{-1} \mathbf{M}_{21} \quad \left(\text{Schur complement of } \mathbf{M}_{22} \right)$$

Consider an $n \times n$ GOE matrix $\mathbf{H}_n$. Let

$$\mathbf{M}_n = z\mathbf{I}_n - \mathbf{H}_n, \quad \text{so} \quad \mathbf{M}_n^{-1} = \mathbf{G}_{\mathbf{H}_n}(z) = \mathbf{G}_n(z).$$

The cavity method compares systems of size n and $n-1$ and exploit that their statistics converge to the same when $n \rightarrow \infty$ to obtain *self-consistent equations* for these statistics (order parameters).

Key Takeaways — Lecture 1

- RMT allows us to analyze the **null** data model (i.e. noise-only)
- Resolvent + Stieltjes transform + inversion formula = **workhorse** toolbox
- Semicircle law describes GOE bulk on $[-2, 2]$
- Next: add a **rank-one spike** (the “signal”) $\Rightarrow$ BBP phase transition and spectral (PCA) estimator

$$\mathbf{Y} = \tau \mathbf{x} \mathbf{x}^T + \mathbf{H}$$

Roadmap (Lecture 2)

- Spiked Wigner model
- BBP phase transition: threshold, outlier location
- Eigenvector overlap (alignment): performance of spectral PCA
- Phase diagram; typical vs worst-case complexity
- Better than PCA: AMP algorithm
- Link with the Sherrington-Kirkpatrick spin glass model

Spiked Wigner Model

Definition (Spiked Wigner model)

Let $\mathbf{H} = \mathbf{H}_n$ be a GOE noise matrix, and $\mathbf{x} \in \mathbb{R}^n$ a fixed (unit) norm “signal”. The observed data is

$$\underbrace{\mathbf{Y}}_{\text{data}} = \tau \underbrace{\mathbf{x}\mathbf{x}^T}_{\text{low-rank signal}} + \underbrace{\mathbf{H}}_{\text{structure-less noise}}$$

$\tau \geq 0$ controls the signal-to-noise ratio.

- PCA viewpoint: does the top eigenpair recover the spike? (RMT scaling with GOE bulk edge at ± 2)

Remark (Extensive rank problem)

- *If the signal matrix has rank $\mathcal{O}(n)$, the problem is much harder.*
- *Spectrum of $\mathbf{Y}$ given by free probability theory / free convolution*
- *Rotational invariant estimator for general noisy matrices (Bun, Allez et al. Trans. Info. Theory 16')*
- *Phase Diagram of Extensive-Rank Symmetric Matrix Denoising beyond Rotational Invariance (Barbier, Camilli et al. Physical Reviews X 25')*

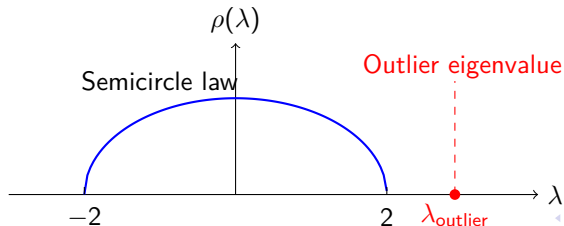
Theorem (Ben Arous-Baik-Péché (BBP) transition)

The largest eigenvalue $\lambda_1 = \lambda_1(\mathbf{Y})$ satisfies:

- If $\tau \leq 1 =: \tau_{\text{BBP}}$, $\lambda_1 \rightarrow 2$ almost surely as $n \rightarrow \infty$
- If $\tau > 1$, $\lambda_1 \rightarrow \tau + \tau^{-1} > 2$ almost surely

Moreover, for $\tau > \tau_{\text{BBP}} = 1$, the eigenvector $\mathbf{v}_1 = \mathbf{v}_1(\mathbf{Y})$ of the data associated with λ_1 almost surely has a non-trivial overlap with the perturbation $\mathbf{x}$:

$$\lim_{n \rightarrow \infty} |\langle \mathbf{v}_1, \mathbf{x} \rangle|^2 = 1 - \tau^{-2} \quad \text{a.s.}$$



Resolvent Analysis: Outlier Location and Critical SNR

Theorem (BBP 1: Outlier Location)

(i) If $\tau \leq 1 =: \tau_{\text{BBP}}$, $\lambda_1(\mathbf{Y}) \rightarrow 2$ as $n \rightarrow \infty$; (ii) If $\tau > 1$, $\lambda_1(\mathbf{Y}) \rightarrow \tau + \tau^{-1} > 2$

Let's prove it.

Theorem (Sherman-Morrison formula)

Given a rank-one matrix $\mathbf{xy}^\top$ of same dimension as the invertible matrix $\mathbf{A}$,

$$(\mathbf{A} + \mathbf{xy}^\top)^{-1} = \mathbf{A}^{-1} - \frac{\mathbf{A}^{-1}\mathbf{xy}^\top\mathbf{A}^{-1}}{1 + \mathbf{y}^\top\mathbf{A}^{-1}\mathbf{x}}$$

- Sherman-Morrison:

$$\mathbf{G}_\mathbf{Y}(z) = (z\mathbf{I}_n - \mathbf{H} - \tau\mathbf{xx}^\top)^{-1} = \mathbf{G}_\mathbf{H}(z) + \frac{\tau\mathbf{G}_\mathbf{H}(z)\mathbf{xx}^\top\mathbf{G}_\mathbf{H}(z)}{1 - \tau\mathbf{x}^\top\mathbf{G}_\mathbf{H}(z)\mathbf{x}}$$

- Outlier iff $\lambda > 2 = \lim_{n \rightarrow \infty} \lambda_1(\mathbf{H})$. For $z > 2$, $\mathbf{G}_\mathbf{H}(z)$ regular. Outlier $\lambda_1 = \lambda_1(\mathbf{Y})$ is thus a pole:

$$1 - \tau\mathbf{x}^\top\mathbf{G}_\mathbf{H}(\lambda_1)\mathbf{x} = 0$$

Resolvent Analysis: Outlier Location and Critical SNR

- In the large n limit, for λ_1 outside $[-2, 2]$, we have by independence of $\mathbf{x}$ and $\mathbf{G}_H(z)$ that

$$\mathbf{x}^T \mathbf{G}_H(\lambda_1) \mathbf{x} \rightarrow g_{\text{sc}}(\lambda_1) \quad (\text{by Hanson-Wright inequality})$$

- Pole equation asymptotically becomes

$$1 - \tau g_{\text{sc}}(\lambda_1) = 0 \implies g_{\text{sc}}(\lambda_1) = \frac{z \pm \sqrt{\lambda_1^2 - 4}}{2} = \tau^{-1}$$

- Solving for λ_1 gives

$$\lambda_1 = \tau + \tau^{-1}$$

- This solution is valid only when $\lambda_1 > 2$ (outside the semicircle support) so that the square root $\sqrt{\lambda_1^2 - 4}$ is real. It implies

$$\tau + \tau^{-1} > 2 \iff \tau > 1$$

- For $\tau \leq 1$, the pole equation has no solution outside $[-2, 2]$, so no outlier appears. The BBP transition where the outlier appears thus happens at

$$\tau = \tau_{\text{BBP}} = 1$$

Performance of Spectral PCA: Eigenvector Overlap

Theorem (BBP 2: Eigenvector Overlap)

For $\tau > \tau_{\text{BBP}} = 1$, $|\langle \mathbf{v}_1(\mathbf{Y}), \mathbf{x} \rangle|^2 \rightarrow 1 - \tau^{-2}$

Let's prove it

- Spectral decomposition of data resolvent (same eigenvectors $(\mathbf{v}_i)$ as $\mathbf{Y} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^\top$),

$$\mathbf{G}_\mathbf{Y}(z) = \mathbf{V} \text{diag}\left(\frac{1}{z - \lambda_i(\mathbf{Y})}\right)_i \mathbf{V}^\top = \sum_{i \leq n} \frac{\mathbf{v}_i \mathbf{v}_i^\top}{z - \lambda_i(\mathbf{Y})}$$

- Thus for $\tau > 1$:

$$|\langle \mathbf{v}_1, \mathbf{x} \rangle|^2 = \lim_{z \rightarrow \lambda_1} (z - \lambda_1) \mathbf{x}^\top \mathbf{G}_\mathbf{Y}(z) \mathbf{x}$$

because only the desired term diverges in this limit.

Performance of Spectral PCA: Eigenvector Overlap

- Sherman-Morrison:

$$\mathbf{G}_Y(z) = \mathbf{G}_H(z) + \frac{\tau \mathbf{G}_H(z) \mathbf{x} \mathbf{x}^\top \mathbf{G}_H(z)}{1 - \tau \mathbf{x}^\top \mathbf{G}_H(z) \mathbf{x}}$$

- We have

$$\lim_{z \rightarrow \lambda_1} (z - \lambda_1) \mathbf{x}^\top \mathbf{G}_Y(z) \mathbf{x} \approx \lim_{z \rightarrow \lambda_1} (z - \lambda_1) \left(g_{\text{sc}}(z) + \tau \frac{g_{\text{sc}}(z)^2}{1 - \tau g_{\text{sc}}(z)} \right).$$

- Using l'Hopital rule to take care of the 0/0 limit, we finally get (using $\tau^{-1} = g_{\text{sc}}(\lambda_1)$)

$$\lim_{n \rightarrow \infty} |\langle \mathbf{v}_1, \mathbf{x} \rangle|^2 = -\frac{g_{\text{sc}}(\lambda_1)^2}{g'_{\text{sc}}(\lambda_1)} = 1 - \tau^{-2}$$

when $\tau > 1$, 0 else (the l.h.s. is non-negative, while the r.h.s. would be negative if $\tau < 1$).

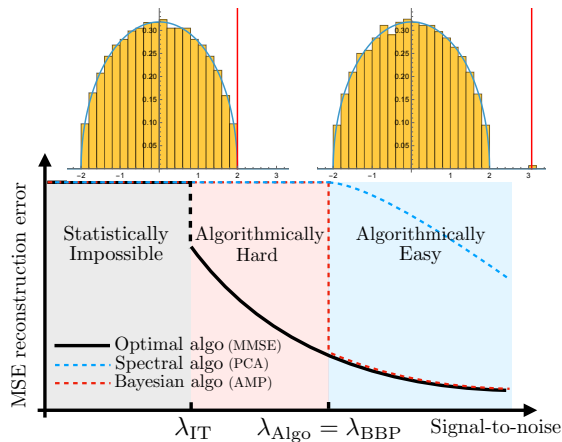
Remark

Spiked covariance: analogous BBP transition with outlier beyond $\lambda_+ = (1 + \sqrt{\gamma})^2$ if spike strong enough.

Key Takeaways on BBP

- Rank-one spike **detaches** once $\tau > \tau_{\text{BBP}} = 1$
- Outlier location: $\lambda_1(\mathbf{Y}) = \tau + \tau^{-1}$
- Eigenvector alignment / spectral PCA performance: $|\langle \mathbf{v}_1(\mathbf{Y}), \mathbf{x} \rangle|^2 \rightarrow 1 - \tau^{-2}$ for $\tau > 1$, 0 else
- Same story for **spiked covariance** relative to MP edge

Spiked-Wigner Model: Phase Diagram



- Double notation (sorry) $\tau = \lambda$
- Regions: *Impossible* / *Hard* / *Easy*, e.g., for sparse prior $P_X(x) = \rho\delta_0 + (1 - \rho)\tilde{P}(x)$
- Thresholds: information-theoretic λ_{IT} , algorithmic λ_{Algo} , spectral (BBP)

Typical vs Worst-Case Complexity

- **Typical-case hardness:** hardness in average over the generative model of the problem instance $\Rightarrow$ sharp thresholds
- **Worst-case hardness:** adversarial inputs; may be much harder to analyze but also less realistic
- Message: gaps (λ_{IT} vs λ_{Algo}) suggest barriers for generic poly-time algorithms

Remark

Whether statistical-to-computational gaps (often delimited by AMP/Message passing algos) are fundamental is the most important question in average-case complexity theory

Beating Spectral PCA: AMP

- **Approximate Message Passing (AMP)**: Bayesian, prior-aware iterative estimator
- **Mean-field analysis**: *state evolution* predicts performance of AMP
- Performance: reaches $\tau_{\text{Algo}} = \lambda_{\text{Algo}}$, often surpasses spectral PCA when the prior is informative

AMP & State Evolution — Setup (Spiked Wigner)

(Rescaled) Rademacher-spiked Wigner:

$$\mathbf{Y} = \frac{\tau}{n} \mathbf{x}^* \mathbf{x}^{*\top} + \frac{1}{\sqrt{n}} \mathbf{W}, \quad \mathbf{x}^* \in \{\pm 1\}^n, \quad \mathbf{W} = \mathbf{W}^\top, \quad (W_{ij}) \sim \mathcal{N}(0, 1 + \delta_{ij}).$$

AMP iteration:

$$\mathbf{v}_{t+1} = \mathbf{Y} \mathbf{x}_t - \underbrace{\langle f'(\mathbf{v}_t) \rangle}_{\text{Onsager term}} \mathbf{x}_{t-1} \quad \mathbf{x}_t = f(\mathbf{v}_t) \quad f(\mathbf{v}) = (\tanh(\tau v_i))_i$$

- Without Onsager: power method + soft nonlinearity; Onsager *cancels* time-correlations from reusing the same W .
- *State evolution* (SE): scalar recursion predicting the law of $\mathbf{v}_t$ in high dimension.

State Evolution (SE) Heuristic

- **Ansatz (decoupling/CLT intuition):** Thanks to Onsager, effective observation is asymptotically distributed as

$$\mathbf{v}_t \approx \mu_t \mathbf{x}^* + \sigma_t \mathbf{g}_t, \quad \mathbf{g}_t \sim \mathcal{N}(0, \mathbf{I}_n) \text{ independent of everything else for each } t$$

Under (heuristic) independence implied by Onsager, and using $f(-v) = -f(v)$,

$$\mathbf{v}_{t+1} = \mathbf{Y}f(\mathbf{v}_t) = \mathbf{x}^* \underbrace{\frac{\tau}{n} \mathbf{x}^{*\top} f(\mathbf{v}_t)}_{\mu_{t+1}} + \underbrace{\frac{1}{\sqrt{n}} \mathbf{W}f(\mathbf{v}_t)}_{\sigma_{t+1} \mathbf{g}_{t+1}} \Rightarrow \text{SE}(n \rightarrow \infty) : \begin{cases} \mu_{t+1} = \tau \mathbb{E}_{\mathbf{g} \sim \mathcal{N}(0,1)} f(\mu_t + \sigma_t \mathbf{g}), \\ \sigma_{t+1}^2 = \mathbb{E}_{\mathbf{g} \sim \mathcal{N}(0,1)} [f(\mu_t + \sigma_t \mathbf{g})^2] \end{cases}$$

- **Prediction:** Iterate SE from small initialisations, yielding $\mathbf{v}_\infty \approx \mu_\infty \mathbf{x}^* + \sigma_\infty \mathbf{g}$
- **Bayes-optimal choice** for $f(\cdot)$ therefore given by the posterior mean

$$f_{\text{opt}}(v) = f_{\text{opt},t}(v) = \mathbb{E}[x|v = \mu_t x^* + \sigma_t g] = \tanh\left(v \frac{\mu_t}{\sigma_t^2}\right)$$

(Only) for this optimal choice $f_{\text{opt},t}$, the “**Nishimori identity**” implies

$$\mathbb{E} f_{\text{opt},t}(\mu_t + \sigma_t \mathbf{g}) = \mathbb{E} [f_{\text{opt},t}(\mu_t + \sigma_t \mathbf{g})^2] \quad \Rightarrow \quad \frac{\mu_t}{\sigma_t^2} = \tau \quad \Rightarrow \quad f_{\text{opt},t}(v) = f_{\text{opt}}(v) = \tanh(v\tau)$$

Consequences & Takeaways

- **Phase transition:** SE shows $\text{MSE of AMP} = 0$ for $\tau \leq 1$ (no correlation), and recovery for $\tau > 1$: same transition as BBP/PCA (in this case of Rademacher prior, but not always)
- **Performance from SE:** any metric (e.g. MSE, overlap, etc) is computable from $(\mu_\infty, \sigma_\infty)$ via $v_\infty \approx \mu_\infty x^* + \sigma_\infty g$.
- **Rigorous status:** with the Onsager term, SE is *provably correct* for AMP on dense models (e.g. Bayati–Montanari '11; Javanmard–Montanari '13).
- **Optimality:** for the Spiked Wigner model, AMP achieves the *information-theoretically optimal* MSE above threshold $\tau_{\text{Algo}} = \lambda_{\text{Algo}}$.
- **Moral:** SE reduces high-dimensional inference to a *scalar denoising recursion*, explaining algorithmic thresholds and guiding design.

Spiked Wigner $\Leftrightarrow$ Planted SK (Intuition)

- **Posterior $\Rightarrow$ Hamiltonian:**

$$P(\mathbf{x}|\mathbf{Y}) \propto \left(\prod_i P_X(x_i) \right) \exp\{-H(\mathbf{x}; \mathbf{Y})\}$$

with negative log-likelihood / Hamiltonian

$$H(\mathbf{x}; \mathbf{Y}) = \sum_{i,j} \left(\frac{\tau^2}{2n} x_i^2 x_j^2 - \tau Y_{ij} x_i x_j \right) \quad \text{with} \quad Y_{ij} = \frac{\tau}{n} x_i^* x_j^* + \frac{1}{\sqrt{n}} W_{ij}$$

- **Binary spins** ($x_i \in \{\pm 1\}$): $x_i^2 = 1$ makes the first term a constant; H reduces to a quadratic form with *random couplings*.
- **Identification with Sherrington-Kirkpatrick spin glass model:**

$$H_{SK}(\boldsymbol{\sigma}; \mathbf{J}) = -\beta \sum_{i,j} J_{ij} \sigma_i \sigma_j \quad \text{with} \quad J_{ij} = \frac{J_0}{n} + \frac{1}{\sqrt{n}} Z_{ij} \quad Z_{ij} \sim \mathcal{N}(0, 1)$$

with a *planted bias* J_0 playing the role of $\mathbf{x}^*$: this is the **planted (teacher-student) SK model**.

- **Nishimori setting:** matched prior/noise $\Rightarrow$ special identities and no RSB in Bayes-optimal inference (Barbier and Panchenko, 19' Communications in Mathematical Physics).

Inference $\leftrightarrow$ Spin Glass: A Quick Dictionary

Statistical mechanics	Bayesian inference
Spins $\sigma_i \in \{-1, 1\}$ / degrees of freedom / particles positions	Variables / bits / unknowns $X_i \in \mathbb{R}$
Quenched disorder J_{ij}	Data / observations y_{ij}
Hamiltonian / energy $H_{\mathbf{J}}(\boldsymbol{\sigma})$	Negative log-posterior $-\log P(\mathbf{x} \mathbf{y})$
Gibbs-Boltzmann distribution $\frac{1}{Z(\mathbf{J})} e^{-\beta H_{\mathbf{J}}(\boldsymbol{\sigma})}$	Posterior distribution $P(\mathbf{x} \mathbf{x}) = \frac{P(\mathbf{x})P(\mathbf{y} \mathbf{x})}{P(\mathbf{y})}$
Partition function $Z_n(\mathbf{J})$	Evidence $P(\mathbf{y})$
Thermal average $\langle f(\boldsymbol{\sigma}) \rangle_{\mathbf{J}}$	Posterior mean $\mathbb{E}[f(\mathbf{x}) \mathbf{y}]$
Disorder expectation $\mathbb{E}_{\mathbf{J}}$	Data expectation $\mathbb{E}_{\mathbf{y}}$
Free energy $F_n = -\frac{1}{\beta n} \ln Z_n$	Shannon entropy $H(\mathbf{y})/n$, mutual information $I(\mathbf{x}; \mathbf{y})/n$.
Control parameter, inverse temperature β	Signal-to-noise ratio λ , relative amount of data n/p
Order parameters, magnetization, overlap	mean-square error, reconstruction error on the signal
Sampling typical/equilibrium configurations	Computing $\hat{\mathbf{x}}_{\text{Bayes}} = \mathbb{E}[\mathbf{x} \mathbf{y}]$
Finding the ground state	Computing $\hat{\mathbf{x}}_{\text{MAP}} = \arg \max P(\mathbf{x} \mathbf{y})$
Replica-symmetry	Concentration of order parameters
Phase transition	Statistical/computational thresholds
Thermodynamic identity $\frac{\partial}{\partial h} F_n = -\langle m_n \rangle$	I-MMSE formula $\frac{\partial}{\partial \lambda} I(\mathbf{x}; \mathbf{y}) = \frac{1}{2} \text{MMSE}(\mathbf{x} \mathbf{y})$